

Appendix

A Dataset Construction

A.1 Fetching Data from Website

Details on CovPDB and CovBinderInPDB datasets. CovPDB contains PDB data collected up to August 31, 2020, encompassing 733 proteins, 1,501 ligands, and 2,294 covalent protein–ligand complexes. The dataset includes 14 targetable residues and 21 covalent reaction mechanisms. In contrast, CovBinderInPDB, which incorporates PDB data up to January 18, 2022, comprises 1,170 protein chains, 2,189 ligands, and 3,555 complex structures, along with 9 targetable residues and 113 reaction patterns. The differences in dataset size and composition reflect variations in their respective data collection strategies.

Data Download and Web Crawling. We obtained CovPDB² and CovBinderInPDB³ dataset from their official website. For CovPDB, since the HET codes of the bound ligands in the PDB files were not provided, we extracted this information by crawling the dataset’s complex list available at https://drug-discovery.vmi.uni-freiburg.de/covpdb/complexes_list/initial=Allsortedby=protein_id. For CovBinderInPDB, since the PDB structure files were not provided, we used the PDB fetch tool [48] to download the corresponding PDB files. We then merged the data from CovPDB and CovBinderInPDB. In cases where entries appeared in both datasets, we prioritized the data from CovPDB due to its higher quality [18]. This merging process resulted in a consolidated dataset of 4,064 examples. To identify covalent bonds, we extracted information from the LINK records in the PDB files. Finally, we retained only those structures that contained exactly one protein–ligand covalent bond.

Gain Pre-reactive Molecule. For CovBinderInPDB, the authors provided the pre-reactive molecule’s SMILES expression in downloaded excel file from their official website. For CovPDB, the authors provided the pre-reactive molecule’s SDF format file. So we additionally used rdkit to get the SMILES expression from SDF file.

A.2 Pre-processing Details

Splitting Complex PDB Files into Protein and Ligand Components. We utilized the bounded ligand’s HET code to identify the target ligand of interest, filtering out all other clashes and water molecules in the PDB file. Furthermore, we exclude entries that have more than one linked residue to the target ligand. We retained only those complex structures whose ligand files, in SDF or MOL2 format, could be processed by RDKit and OpenBabel.

Select Protein Chains and Pocket Atoms. Our pocket radius is set to 10 Å. To prevent the input residue number for the reactive location prediction model from being too long, we filter out chains where the minimum contact distance between amino acid atoms and ligand atoms exceeds the 10 Å cutoff. We retained only those proteins that have an atom within a 10 Å radius of any ligand atom. We also filtered out complexes with alternate positions for ligands or bonded amino acids, retaining only one binding scenario for symmetric ligands and receptors. Furthermore, we excluded complexes that still contain more than 1,022 residues after the

aforementioned filtering process. We set 1,022 as the maximum length to reserve two token positions for beginning-of-sequence (BOS) and end-of-sequence (EOS) tokens.

Gain Post-reactive Molecule. The CONNECT field specifies connectivity but does not distinguish between single and double bonds. Moreover, since hydrogen atoms are not provided in the PDB files, it is impossible to deduce the corresponding connections through the valency of the atoms. Thus, we adopted a workflow similar to that of CovBinderInPDB to derive the post-reactive molecules. First, we constructed SDF files based on the mmCIF files of the stable form of the bound ligand when isolated, as provided in the Chemical Component Dictionary⁴. We then compared these with the structure of the post-reactive ligand in PDB file, removing extra atoms from the constructed SDF files. This includes removing extra non-hydrogen atoms in the Chemical Component Dictionary and one hydrogen atom connected to the bonded ligand atom. Subsequently, we connected corresponding parts of the amino acids to these structures. We then applied patches for several special cases: converting double/triple bonds to single/double bonds and adding a hydrogen atom. Additionally, we marked the boron in the boronation reactions with a negative charge. This post-reactive molecule process is applied to the data from both CovPDB and CovBinderInPDB to ensure consistency. We retained only those ligands that could be aligned between the Chemical Component Dictionary and the PDB file.

A.3 Statistics on the Crafted Datasets

The statistical results of amino acid types in our dataset are illustrated in Figure 6 and Figure 8. Our analysis reveals a diverse distribution of reactive mechanisms, including Michael addition, ring opening, and others, highlighting the comprehensive coverage of binding mechanisms in our benchmark. Moreover, the distribution of target amino acid types significantly exceeds that of previous works. As shown in Figure 7 and Figure 9, the reactive amino acid types are far more extensive compared to earlier studies primarily focused on Cys or Ser [50, 60].

B Modeling for Each Task

All our experiments were run on 8 NVIDIA V100 16GB GPUs.

B.1 Reactive location Prediction

Model Details. Our implemented reactive site model can be viewed in Figure 3. The ligand encoder is finetuned from Uni-Mol pre-trained checkpoint on 209M molecular conformations. The protein encoder is randomly initialized since our protein encoder is a residue-level model, whereas Uni-Mol’s pocket model is an atom-level model. In order to merge features from both protein and ligand, we utilize one cross-attention layer, where the query is from the protein, and the key and value are from the ligand, to get the output of the same length as the protein. We refer to the cross-attention implementation of Stable Diffusion in [57]. The output complex embedding from the cross-attention layer is passed through a sigmoid layer to get the weight value of the current amino acid. The final pocket center is a weighted average of all input amino acids’

²<https://drug-discovery.vmi.uni-freiburg.de/covpdb/download>

³<https://yzhang.hpc.nyu.edu/CovBinderInPDB/>

⁴Here we used Ligand Expo[14].

C_α coordinates as $\mathcal{P}^c = \frac{\sum_j p_j \times x_j}{\sum_j p_j}$ where p_j is the predicted pocket probability for j -th amino acid.

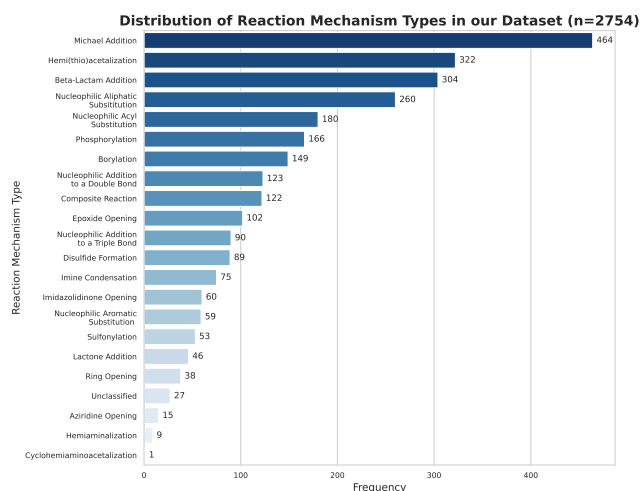


Figure 6: Reactive Mechanism Type Statistics (n=2754)

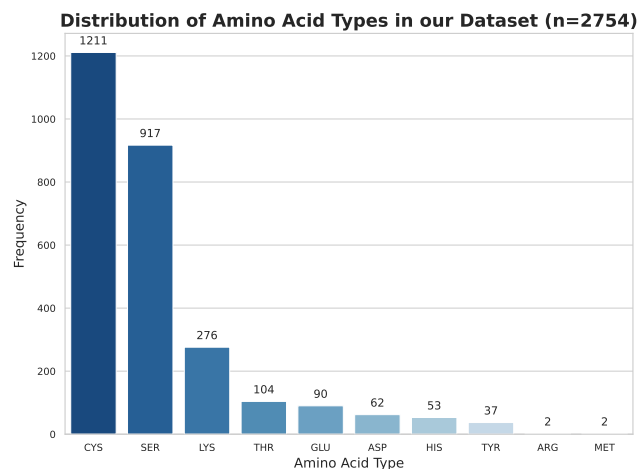


Figure 7: Reactive Amino Acid Type Statistics (n=2754)

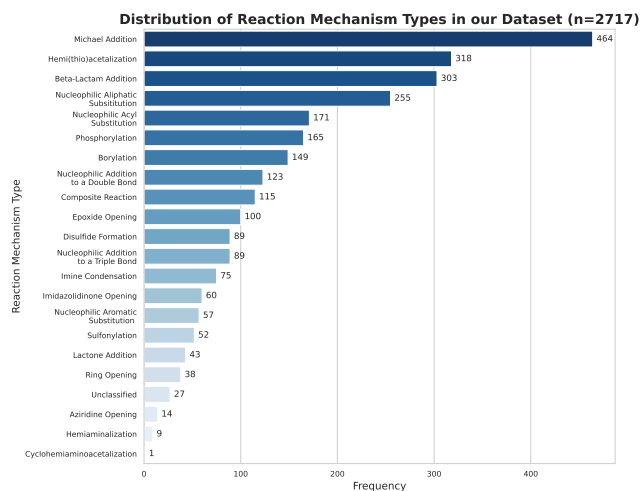


Figure 8: Reactive Mechanism Type Statistics (n=2717)

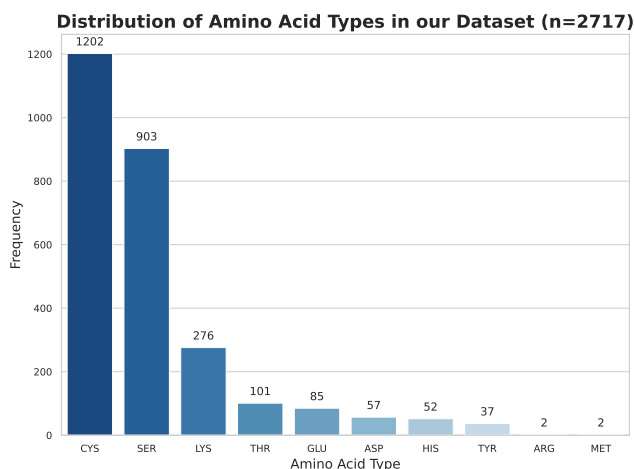


Figure 9: Reactive Amino Acid Type Statistics (n=2717)

Hyperparameters. Huber loss [23] is used as the loss function for the pocket prediction subtask. During the inference stage, the pocket residues are identified by the 20Å distance threshold from the predicted pocket center. For the reactive amino acid prediction subtask, the model output is a sequence-level classification result based on the pocket residues' complex embedding from the cross-attention layer's output, and cross-entropy loss is used as the loss function. Instead of directly using the model-predicted pocket residues as input for the reactive amino acid prediction subtask during training, which disobeys the teacher forcing style training [19], we use the ground truth pocket residues during training and the predicted pocket residues during inference. Adam [27] is used as the optimizer when training our model and betas in Adam are 0.9 and 0.99 for all experiments. The protein model consists of 25 layers of Uni-Mol blocks. Loss weight α for reactive amino acid prediction subtask is 0.05 for our model.

B.2 Covalent Reaction Prediction

Developed in 2022, T5Chem utilizes the Text-to-Text Transfer Transformer (T5) framework [46] and self-supervised pretraining with PubChem molecules for diverse chemical reaction predictions, including reaction type classification and synthesis predictions. ReactionT5, introduced in 2023, extends this model by pretraining on the Open Reaction Database (ORD) to improve yield and product predictions. Chemformer also leverages a Transformer-based architecture and self-supervised pretraining to achieve top-tier accuracy in synthesis tasks. In our implementation, we used the default hyperparameters and model settings for each model.

Table 8: An overview on current covalent docking tools.

Method	Publish year	Open-sourced code	Public accessible
GOLD (cov) [56]	2003	✗	✗
ICM-PRO [26]	2007	✗	✗
FITTED (cov) [37]	2012	✗	✗
CovalentDock [43]	2013	✓	✓
CovDock [66]	2014	✗	✗
DOCKoalent [34]	2014	✗	✗
DOCKTITE [52]	2015	✓	✗
AutoDock4 (cov) [5]	2016	✓	✓
MOE (cov) [6]	2018	✗	✗
WIDOCK [51]	2021	✗	✗
COV_DOX [58]	2022	✗	✗
CDOCKER (cov) [62]	2022	✓	✓
HCovDock [61]	2023	✗	✓
AC (cov) [18]	2023	✗	✓

B.3 Covalent Docking Pose Prediction

Ligand and Pocket Preparation for Noncovalent Docking. To obtain the input ligand for noncovalent models from the output of the reaction model, which contains both the bonded amino acid part and the bonded ligand part, we further process the post-reactive ligand SMILES as follows. First, we remove the amino acid parts from the post-reactive ligand’s SMILES. Next, we align this SMILES with the corresponding post-reactive ligand’s PDB file to synchronize the bond order and coordinate information. Finally, we employed ETKDG [47] algorithm using RDKit [32] to randomly generate a low-energy conformation as the initial pose of the ligand. To obtain the input pocket for docking models, we used OpenBabel [42] to obtain coordinates for all pocket atoms, following the same criterion for pocket atom identification as in reactive site prediction task.

Loss Function. We employed $\mathcal{L}_{dist}$ for the same as Uni-Mol to use $\mathcal{L}_{dist} = \text{SmoothL1Loss}(D_{intra}, D'_{intra}) + \text{MSELoss}(D_{inter}, D'_{inter})$ as the main loss, where D_{intra} is the distance map between ligand atoms and ligand atoms, while D_{inter} is the distance map between ligand atoms and pocket atoms. The prime version denotes the target values. The auxiliary $\mathcal{L}_{cov}$ loss is demonstrated in Equation 5.

Postprocessing Details. If the covalent inter-bond length between the bonded ligand atom and the bonded pocket atom deviates more than 10σ from the mean covalent inter-bond length (where σ is the standard deviation), the bonded ligand atom will be moved to a new coordinate to ensure the covalent inter-bond length satisfy the normal distribution.

Hyperparameters. We used the default hyperparameters as Uni-Mol docking model. We experimented with adjusting the loss weight for $\mathcal{L}_{cov}$, but initial experiments indicated that maintaining the loss weight at 1 yielded better results.

C Baselines for Reactive Location Prediction

P2Rank, an open-source software available as both a standalone command line program and a Java library, features a user-friendly, lightweight installation and does not require other bioinformatics tools or extensive databases. It is noted for rapid processing and automated predictions, making it ideal for large datasets and scalable bioinformatics pipelines. Conversely, Fpocket employs a fast geometric method involving the filtering and clustering of alpha spheres via Voronoi tessellation. This tool is favored for its extensive use in large-scale projects and its ability to generate a high

number of predicted pockets per protein. We used the default parameter for each tool. To obtain the final pocket center coordinate, for P2Rank, we chose the rank 1 predicted pocket center result. For Fpocket, since it does not output a pocket center coordinate, we chose the rank 1 predicted atoms contacted by alpha spheres in the given pocket and take the geometric center of these atoms as the predicted pocket center to calculate the DCC metric.

D Baselines for Traditional Covalent Docking Tools

A short review on current covalent docking tools can be seen in Table 8. The check in the "Publicly Accessible" column indicates that either their code is publicly accessible or they have published a website server. DOCKoalent’s website is still accessible, but the docking service is already down. DOCKTITE is written as an SVL script for MOE. Even though DOCKTITE open-sourced their script code, MOE is not publicly accessible, so DOCKTITE is still not publicly accessible.

When compared to traditional covalent docking tools, we excluded closed-source and web server-based tools due to their limited accessibility and lack of transparency, which are essential for reproducible benchmarks. This decision ensures our benchmark remains robust, accessible, and adaptable for in-depth evaluation. Due to significant challenges with the runtime environments of CovalentDock and CDOCKER, as well as the fact that CovalentDock is outdated and no longer maintained, we were unable to successfully run these tools. As a result, AutoDock4 (cov) was the only traditional covalent docking tool used as a baseline in our experiment.

AutoDock4 (cov) Baseline. Two different covalent docking procedures were implemented in AutoDock4 (cov) [5]. We chose the flexible side chain method as it performed better [5]. The use of this method requires the addition of two link atoms from the target residue to the ligand. So the input is further processed based on the input to non-covalent docking model to keep the additional two atoms from the post-reactive ligand. Close attention must be paid to the geometry of the initial modification, or the docking process may fail. The following steps are the same as their default processing workflow. During the docking, the protein and the backbone of its reactive site are treated as rigid and the $C\alpha$ and $C\beta$ atoms are fixed. The ligand is considered a fully flexible extension of the protein reactive site.

E Analysis on the Minimum Distance in Inter Distance Map

To demonstrate the rationale of our covalent auxiliary loss $\mathcal{L}_{cov}$, we performed a statistical analysis on the relationship between the minimum distances in the inter distance map and the covalent inter-bond distances.

To test whether the minimum distances in the inter distance map and covalent inter-bond distances are drawn from the same distribution, we performed a Kolmogorov-Smirnov test [22] on 2754 data points. The resulting p-value is 0.9999999. The p-value is greatly above the usual confidence level threshold of 0.05, so we cannot reject the null hypothesis that the two samples were

Table 9: Results for cov-docking task after reordering

Methods	% RMSD Below $\uparrow$				% RMSD (IB) Below $\uparrow$			
	2Å	3Å	4Å	5Å	0.5Å	1Å	2Å	3Å
<i>Traditional methods</i>								
AUTOdock4 (cov)	5.2% $\pm$ 0.3%	20.3% $\pm$ 2.2%	36.9% $\pm$ 2.5%	50.7% $\pm$ 3.2%	-	-	-	-
VINA	7.9% $\pm$ 0.9%	23.5% $\pm$ 1.4%	40.4% $\pm$ 0.8%	55.3% $\pm$ 1.1%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	7.0% $\pm$ 0.3%	26.5% $\pm$ 3.6%
SMINA	13.9% $\pm$ 0.8%	35.0% $\pm$ 1.6%	57.1% $\pm$ 0.7%	75.8% $\pm$ 1.2%	0.0% $\pm$ 0.0%	0.6% $\pm$ 0.3%	12.9% $\pm$ 1.4%	39.2% $\pm$ 1.1%
<i>Deep learning-based methods</i>								
UNI-MOL	40.1% $\pm$ 4.4%	59.5% $\pm$ 1.0%	79.2% $\pm$ 0.7%	89.5% $\pm$ 1.1%	59.9% $\pm$ 3.1%	72.6% $\pm$ 2.0%	78.8% $\pm$ 1.6%	83.7% $\pm$ 2.8%
OURS	42.0% $\pm$ 0.7%	60.7% $\pm$ 1.4%	79.2% $\pm$ 3.2%	90.4% $\pm$ 0.7%	60.1% $\pm$ 2.0%	73.1% $\pm$ 2.5%	78.3% $\pm$ 2.2%	83.6% $\pm$ 1.3%
OURS-p	42.0% $\pm$ 0.7%	61.0% $\pm$ 1.2%	79.8% $\pm$ 3.9%	90.6% $\pm$ 0.4%	79.1% $\pm$ 0.7%	93.1% $\pm$ 0.7%	98.4% $\pm$ 0.7%	100.0% $\pm$ 0.0%

drawn from the same distribution. Furthermore, the frequency of covalent inter-bond distances being the minimum distance in the inter distance map is 0.996. This frequency observation also supports our hypothesis that covalent inter-bond distances should be the minimum distances in the inter distance map.

F Results of Covalent Docking after Reordering using Hungarian Algorithm

Table 9 shows the ligand RMSD results after reordering using Hungarian algorithm. From the table we can see that AutoDock4 (cov) is actually the worst.

G Limitation

Covalent docking presents significant challenges compared to traditional non-covalent docking, particularly in problem modeling and

algorithm design, leading to poor performance of current methods on such tasks. Our study has made meaningful progress in covalent docking methodologies. However, we keep only one binding scenario for symmetric ligands and receptors, which may introduce significant discrepancies compared to real data, particularly in blind covalent docking scenarios. Additionally, our approach does not adequately model internal covalent bonds within ligands or between pocket atoms. Our current setting uses a flexible ligand and a rigid pocket, which limits the real-world applicability of our results, as most real-world scenarios involve flexible pockets. Future research should focus on developing more integrated, end-to-end solutions to reduce these cumulative errors and enhance the flexibility of both ligands and pockets to better mimic real biochemical interactions.